



Saint Columban College
College of Computing Studies
Pagadian City



Super SSH

Super SSH



Easy

General Skills

picoCTF 2024

shell

ssh

browser_webshell_solvable

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Description

Using a Secure Shell (SSH) is going to be pretty important.
Additional details will be available after launching your
challenge instance.

This challenge launches an
instance on demand.

Its current status is:

NOT_RUNNING

Launch Instance

Hints ?

1 2 3 4

81,663 users solved



94%
Liked



Author: The Analyst: Hyposelenia

Challenge: Super SSH

Category: General Skills

Date: 01/14/26



I. Objective

The objective of this challenge was to securely connect to a remote server using **SSH (Secure Shell)** and retrieve the flag provided upon successful authentication.

II. Background

Secure Shell (SSH) is a cryptographic network protocol used to securely access remote systems over an unsecured network. Unlike plain text connections, SSH provides **encryption, authentication, and data integrity**, making it a standard tool for remote administration and secure communication.

In this challenge, the hostname, port number, username, and password are dynamically generated and provided only after launching the challenge instance.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – Used as the Linux environment
- **Linux Terminal** – Used to execute decoding commands
- **SSH (OpenSSH client)** – Used to establish a secure remote connection

IV. Methodology

1. The challenge instance was launched to obtain the **hostname, port, username, and password**.
2. Based on the challenge title and description, SSH was identified as the correct protocol.
3. I opened my VirtualBox – Kali Linux. You can simply use the webshell if you're taking this challenge on picoCTF
4. The SSH connection was established using the following command format:

Ssh username@hostname -p port



5. The host authenticity prompt was accepted.
6. Typed “yes” to reassure connection
7. Input the provided password when asked. **(note that you can’t see what you typed in password, that’s just how secure it is)**
8. Upon successful authentication, the server displayed the flag and closed the connection.

V. Result

picoCTF{s3cur3_c0nn3ct10n_45a48857}

```
(kali㉿kali)-[~]
$ ssh ctf-player@titan.picoctf.net -p 49919
The authenticity of host '[titan.picoctf.net]:49919 ([3.139.174.234]:49919)' can't be established.
ED25519 key fingerprint is SHA256:4S9EbTSSRZm32I+cdM5TyztHpQryv5kudRP9PIKT7XQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[titan.picoctf.net]:49919' (ED25519) to the list of known hosts.
ctf-player@titan.picoctf.net's password:
Welcome ctf-player, here's your flag: picoCTF{s3cur3_c0nn3ct10n_45a48857}
Connection to titan.picoctf.net closed.

(kali㉿kali)-[~]
$
```

VI. Explanation

The key concept in this challenge is understanding the **protocols**:

- **SSH** is a full protocol that requires encryption, authentication, and key exchange.

Attempting to use nc on an SSH service fails because SSH expects a secure handshake rather than plain text input. Additionally, specifying the port correctly using the -p flag is essential, as SSH defaults to port 22 if no port is provided.

This challenge reinforces the importance of selecting the correct tool based on the service protocol rather than relying solely on a host and port combination.



VII. Conclusion

This challenge demonstrated the proper use of SSH for secure remote access and emphasized protocol awareness in cybersecurity. Recognizing when to use SSH instead of generic networking tools is a critical skill, especially in real-world systems administration and capture-the-flag competitions.

Successfully completing this challenge required understanding SSH syntax, authentication flow, and the importance of encrypted connections.

— **The Analyst: Hyposelenia**